

Multi-region CL inflow fits of the SPARC category-D galaxies: can the structured residuals be absorbed by the existing model extensions?

Automated analysis for E.P.J. de Haas

Summary

Category D contains the 18 SPARC galaxies whose single-Lagrangian (single-L) residuals are significantly structured (runs test $p < 0.05$) but match neither the $+-+$ template of category B nor the $-+-$ template of category C. Before attributing their kinematics to the history of the galaxy (mergers, accretion), we test whether the existing CL model extensions absorb the structure: one virial window [1,2], two Lagrangians [2], two virial windows [1], and two Lagrangians with a disk window. The result is a split into four groups, not a single class. (i) In 6 galaxies no extension is justified ($\Delta\text{BIC} \geq -6$): the sign pattern is real, but its amplitude lies within the SPARC errors (single-L $\chi_\nu^2 = 0.03\text{--}0.26$). (ii) 3 galaxies are described cleanly by one of the extensions with well-determined parameters. (iii) 4 are improved, but their parameters are poorly constrained. (iv) 5 bulge-dominated S0–Sbc galaxies are improved enormously (ΔBIC down to -3916), but only with unphysical parameters: the inner CL bulge collapses ($R \rightarrow 0$) and window strengths reach $|p| > 100$. For this group the structure points to a missing bulge component rather than to history. Category D therefore does not support the general conclusion that historical dynamics dominate. It is mainly a mixture of residuals within the errors, curves needing more regions, and massive bulges not represented by the CL bulge.

1 Models

All models use the CL inflow quantities $X = \sqrt{2GM/R} - H_z R$ and $Y(r) = \sqrt{2GM/r} - H_z r$ with $H_z = 2.2 \times 10^{-18} \text{ s}^{-1}$ and are fitted to v^2 with $\sigma_{v^2} = 2V_{\text{obs}}\delta V_{\text{obs}}$.

- **S** single-L [2, Eqs. 17–18]: $v^2 = \frac{1}{2}X^2r^2/R^2$ ($r \leq R$), $\frac{3}{2}X^2 - Y^2$ ($r > R$); $k = 2$.
- **VW** one virial window [1, Eq. 6; 2, Eq. 19]: $+p[\frac{1}{2}Y(r)^2 - \Phi_p]$ for $r \geq r_v$; $k = 4$.
- **2L** two Lagrangians [2, Eqs. 24–26]: $L_1(M_1, R_1)$ up to R_2 , $L_2(M_2, R_2)$ beyond; $k = 4$.
- **VW2** two virial windows [1, Eqs. 5–7]: strength p on $[r_{v1}, r_{v2})$, q beyond r_{v2} ; $k = 6$.
- **2LVW** two Lagrangians with a window of strength p in the L_2 disk beyond $r_v > R_2$; $k = 6$.

Window gauges Φ are fixed by continuity of v^2 at each window radius, and window strengths are fitted as descent masses $M_N = (p - 2)M$ (see the category-C report). For $H_z \rightarrow 0$ a window adds GM_N/r plus a constant, so $p > 2$ is a Keplerian descent, $0 < p < 2$ a flattening and $p < 0$ an extra rise.

Selection. The preferred model has the lowest BIC, with a simpler model preferred when it lies within $\Delta\text{BIC} < 2$. An extension is accepted only if it beats single-L by $\Delta\text{BIC} < -6$. For galaxies with $n \leq 7$ the six-parameter models are not fitted.

Status of the preferred fit. **Clean:** all parameters within 60%, no parameter at a bound, $|p|, |q| \leq 100$, $\chi_\nu^2 \leq 2$. **Weakly constrained:** some parameter uncertainty above 60% (degenerate radii or masses), or $\chi_\nu^2 > 2$. **Phenomenological:** the bulge radius at its lower bound ($R \leq 0.025 \text{ kpc}$) or a window strength $|p| > 100$; the fit is statistically excellent but the parameters have no physical CL reading. Uncertainties: covariance, leave-one-out jackknife, distance/inclination refits and an $H_z = 0$ refit, as in the A–C reports.

Table 1: Category-D outcome by status of the preferred fit.

Status	N	Galaxies (type, preferred model)
single-L adequate	6	DDO064 (Im, single-L), F563-1 (Sm, single-L), NGC0024 (Sc, single-L), NGC2915 (BCD, single-L), PGC51017 (BCD, single-L), UGC01281 (Sdm, single-L)
clean	3	NGC0300 (Sd, one window), NGC2841 (Sb, two windows), NGC6015 (Scd, two-L + disk window)
weakly constrained	4	NGC1090 (Sbc, two-L + disk window), NGC4138 (S0, one window), NGC7814 (Sab, one window), UGC00128 (Sdm, two windows)
phenomenological	5	NGC0289 (Sbc, two windows), UGC02487 (S0, two windows), UGC03546 (Sa, two windows), UGC06786 (S0, two windows), UGC09133 (Sab, two windows)

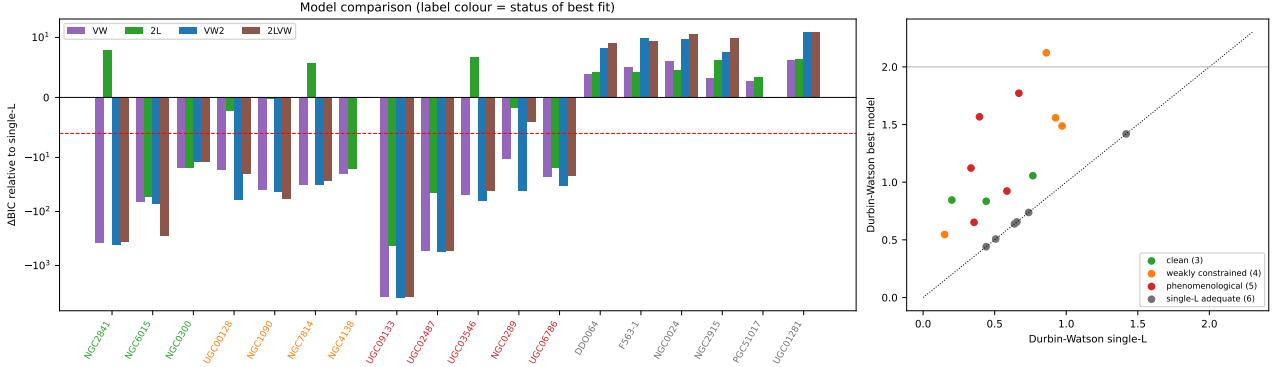


Figure 1: Left: ΔBIC of each extension relative to single-L (symlog scale; red: -6). Galaxy labels are coloured by the status of the preferred fit: clean (green), weakly constrained (orange), phenomenological (red), single-L adequate (grey). Right: Durbin–Watson statistic, single-L versus preferred model.

2 Outcome

Single-L adequate (6). DDO064, F563-1, NGC0024, NGC2915, PGC51017, UGC01281. These are mostly late types and dwarfs (Im, Sm, Sc, BCD, BCD, Sdm) with very small single-L χ^2_ν (median 0.24). The runs test detects a coherent sign pattern, but no extension improves BIC, because the pattern lies well inside the error bars. UGC 1281 ($\chi^2_\nu = 0.03$) is the galaxy for which [1] introduced the virial term on the basis of its outer six points. With the full error bars and BIC this refinement is not statistically required; [1] reported the same, a reduction of RMS_{rel} from 6.5% to 6.3%.

Clean (3). NGC 2841 (Sb) is described by two windows: a Keplerian descent ($p \approx 14$) beyond $r_{v1} \approx 13$ kpc, followed by a renewed rise ($q < 0$) beyond $r_{v2} \approx 44$ kpc. χ^2_ν falls from 9.3 to 0.40 and $\Delta\text{BIC} = -413$. This is a category-C descent with an outer rise, exactly the two-window configuration of [1]. NGC 6015 (Scd) is described by a Lagrangian reset at $R_2 \approx 2.3$ kpc followed by a flattening window ($p \approx 1.8$) at $r_v \approx 3.5$ kpc, i.e. a B-type and a C-type feature combined (χ^2_ν 8.7→1.7). NGC 300 (Sd) needs one rising window ($p < 0$) at $r_v \approx 2.5$ kpc. Two-L is equally good here ($\Delta\text{BIC} +0.05$), so NGC 300 is a B/C borderline case.

Weakly constrained (4). NGC 7814 (Sab) is a descent from the first point; the window reproduces it (χ^2_ν 2.96→0.67), but R and r_v (0.4 and 0.9 kpc) are degenerate with the inner masses. NGC 4138 (S0) has only 7 points. NGC 1090 (Sbc) prefers a two-L + window fit in which R_2 collapses onto R_1 , i.e. effectively one window with a mass change. UGC 128 (Sdm) is improved by two windows (χ^2_ν 6.1→2.9) but remains structured. These galaxies are compatible with the C or B pictures, but the data do not fix the parameters.

Phenomenological (5). NGC0289, UGC02487, UGC03546, UGC06786, UGC09133 (Sbc, S0, Sa, S0, Sab). Two virial windows lower χ^2_ν dramatically (UGC 9133: 64→4.7; UGC 2487: 38→0.66), but in every case the parameters become extreme: the inner CL bulge shrinks to $R \leq 0.06$ kpc (four of five) and window strengths reach $|p|$ or $|q| \approx 10^2$ – 3×10^3 . The windows then act as a flexible spline, not as virial regions. All five are early and intermediate types (S0–Sbc) with prominent bulges: they have a sharp inner peak that the parabolic CL bulge ($v^2 \propto r^2$ inside R) cannot follow. The structure in their residuals is therefore a model limitation of the bulge description and should not be read as evidence of history.

3 Fit quality of the accepted extensions

Table 2: Fit-quality statistics (medians, single-L $\rightarrow$ preferred model).

	Clean (3)	Weakly constrained (4)	Phenomenological (5)
χ^2_ν	8.66 $\rightarrow$ 0.40	4.27 $\rightarrow$ 0.60	3.40 $\rightarrow$ 0.73
RMS _{rel}	0.140 $\rightarrow$ 0.064	0.154 $\rightarrow$ 0.075	0.175 $\rightarrow$ 0.035
Durbin–Watson	0.44 $\rightarrow$ 0.85	0.89 $\rightarrow$ 1.52	0.39 $\rightarrow$ 1.12
$p_{\text{runs}} < 0.05$ after	3/3	1/4	4/5
ΔBIC (median)	-286	-45	-63

Residual structure remains (runs $p < 0.05$) in 8 of the 12 accepted fits, including NGC 2841 and NGC 6015. These are high-precision curves ($Q=1-2$, $n = 44-50$) in which small coherent deviations, e.g. from spiral-arm streaming, become significant at the few-per-cent level (RMS_{rel} 0.02–0.06). As in categories A–C, the SPARC errors are conservative for most galaxies. Distance and inclination systematics are of the same order as the statistical errors. Setting $H_z = 0$ changes the clean fits by at most 7%; the phenomenological fits move by up to about 110%, consistent with their degeneracy.

4 Assessment: history versus model structure

Can the dynamics of category D be attributed to the history of the galaxies? Not on the basis of these fits. Step 1 shows that the class is heterogeneous:

1. **6 galaxies** have structure within the errors; single-L is statistically sufficient.
2. **7 galaxies** are absorbed, cleanly or with weak constraints, by the existing extensions of [1,2]: descent and rising windows, a Lagrangian reset, or combinations of these. They are B- and C-type features that occur together or close to the centre, which is why the single-template classification missed them.
3. **5 bulge-dominated S0–Sbc galaxies** require unphysical parameters. Their residuals trace a massive, compact bulge that the CL bulge parabola cannot represent, which is a limitation of the inner model.

Galaxy history (accretion, mergers, settled virial matter) remains the interpretation proposed in [1] for virial windows in general. That applies equally to categories C and D, and is not established for D specifically. A history-specific conclusion would require step 2: comparing the galaxies that still need windows with independent indicators of disturbance, such as lopsided or tidal morphology, approaching/receding asymmetry of the rotation curve, HI warps and environment. For the phenomenological group, a separate bulge component is needed first: either a Newtonian or CL bulge with its own scale, or the inner bulge region of [1] with a free inner profile. Only then can its residual structure be interpreted.

References

- [1] E.P.J. de Haas (2026), From Rotation Curves to Cosmic Time: Probing High-Redshift Expansion through Nested Spiral Dynamics, *J. High Energy Phys. Grav. Cosmol.* 12, 334–367, doi:10.4236/jhepgc.2026.121022 (virial term Eq. 3; multi-region virial model Eqs. 5–7).
- [2] E.P.J. de Haas, Galactic Rotation Curves and the Constant–Lagrangian Field: Empirical Tests within the Q_g Rotor Framework (manuscript; single-L Eqs. 17–18, virial term Eq. 19, two-L Eqs. 24–26).
- [3] F. Lelli, S.S. McGaugh, J.M. Schombert (2016), SPARC, *AJ* 152, 157.

Table 3: All category-D galaxies: residual signs of the single-L fit and model comparison (ΔBIC relative to single-L; $-$ = not fitted, $n \leq 7$).

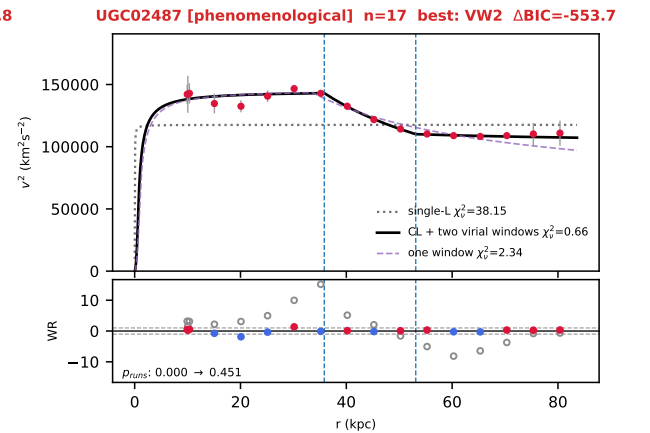
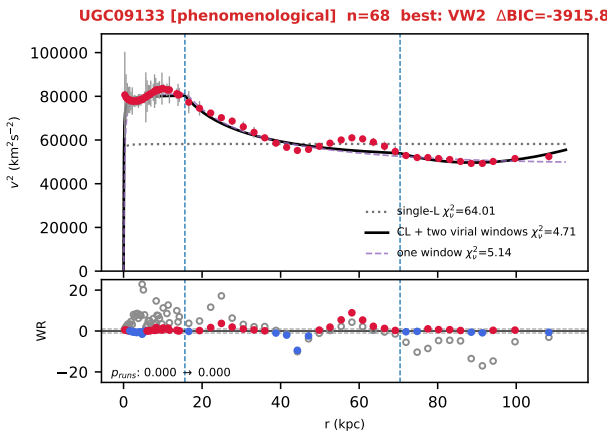
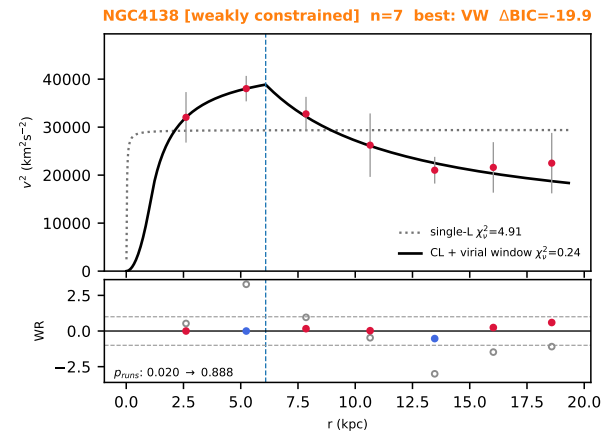
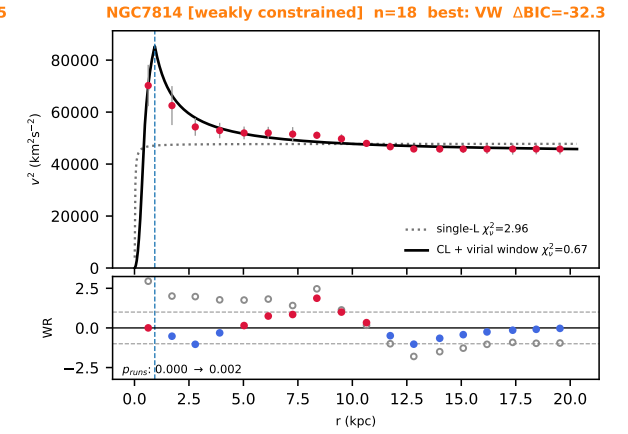
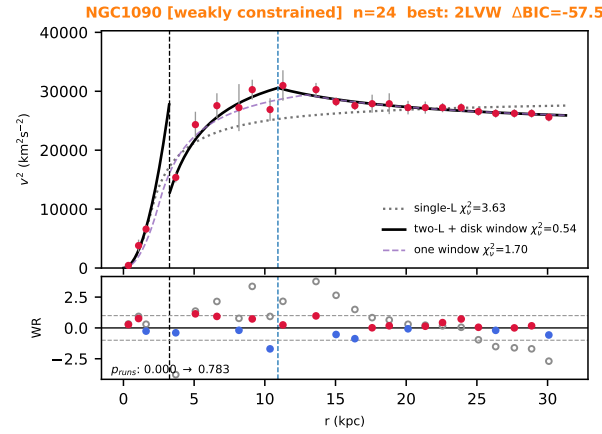
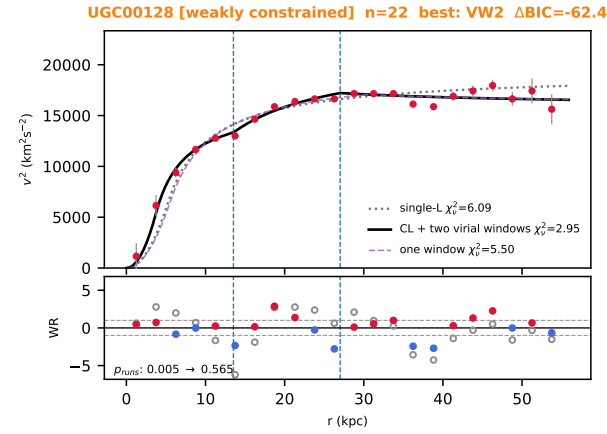
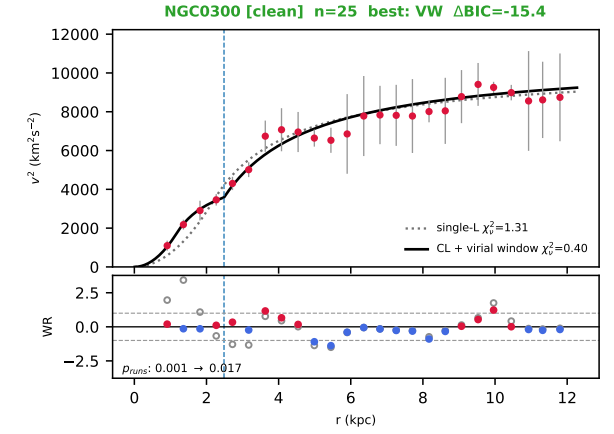
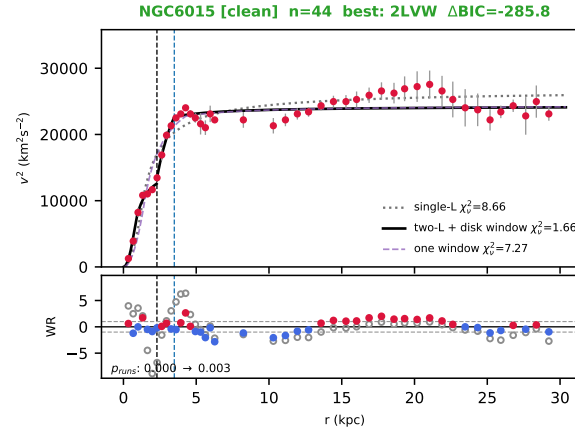
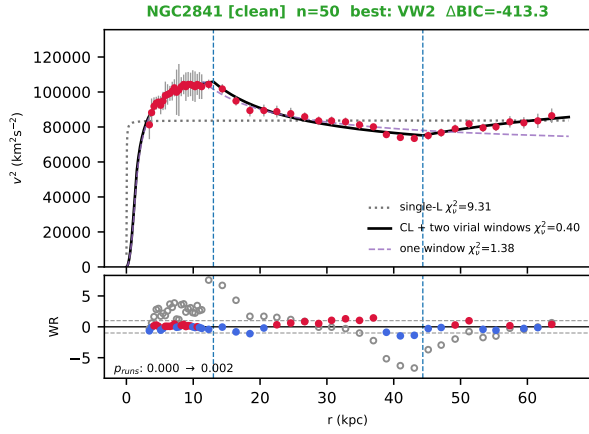
Galaxy	Type	Q	n	single-L residual signs	$\chi^2_{\nu,S}$	p_{runs}	VW	2L	VW2	2LVW	best	$\chi^2_{\nu,\text{best}}$	$p_{\text{runs,best}}$	Status
NGC2841	Sb	1	50	-----	9.31	2.4e-10	-375.5	7.8	-413.3	-367.7	VW2	0.40	0.002	clean
NGC6015	Scd	2	44	++++	8.66	2.6e-06	-65.4	-54.6	-73.9	-285.8	2LVW	1.66	0.003	clean
NGC0300	Sd	2	25	++++	1.31	1.4e-03	-15.4	-15.3	-12.0	-12.0	VW	0.40	0.017	clean
UGC00128	Sdm	1	22	++++	6.09	4.6e-03	-16.8	-2.2	-62.4	-20.6	VW2	2.95	0.565	weakly constrained
NGC1090	Sbc	1	24	++++	3.63	3.5e-04	-39.6	-0.2	-44.2	-57.5	2LVW	0.54	0.783	weakly constrained
NGC7814	Sab	1	18	++++	2.96	5.1e-05	-32.3	5.8	-32.9	-26.7	VW	0.67	0.002	weakly constrained
NGC4138	S0	2	7	++++	4.91	2.0e-02	-19.9	-16.3	-	-	VW	0.24	0.888	weakly constrained
UGC09133	Sab	1	68	++++	64.01	8.6e-14	-3887.2	-437.5	-3915.8	-3878.7	VW2	4.71	0.000	phenomenological
UGC02487	S0	1	17	++++	38.15	8.9e-05	-536.2	-44.7	-553.7	-532.2	VW2	0.66	0.451	phenomenological
UGC03546	Sa	1	30	++++	3.40	1.7e-04	-49.1	6.8	-62.6	-41.6	VW2	0.80	0.013	phenomenological
NGC0289	Sbc	2	28	++++	2.70	1.2e-03	-10.8	-1.8	-40.9	-4.1	VW2	0.73	0.028	phenomenological
UGC06786	S0	1	45	++++	1.41	3.5e-04	-23.4	-15.9	-33.2	-21.9	VW2	0.31	0.001	phenomenological
DD0064	Im	1	14	++++	0.24	1.4e-02	3.9	4.3	8.3	9.0	S	0.24	0.014	single-L adequate
F563-1	Sm	1	17	++++	0.24	1.2e-02	5.1	4.2	9.8	9.3	S	0.24	0.012	single-L adequate
NGC0024	Sc	1	29	++++	0.24	2.4e-02	6.1	4.6	9.7	11.3	S	0.24	0.024	single-L adequate
NGC2915	BCD	2	30	++++	0.26	1.5e-03	3.3	6.3	7.6	9.9	S	0.26	0.001	single-L adequate
PGC51017	BCD	3	6	++++	0.21	3.4e-02	2.8	3.5	-	-	S	0.21	0.034	single-L adequate
UGC01281	Sdm	1	25	++++	0.03	2.6e-04	6.2	6.4	12.3	12.4	S	0.03	0.000	single-L adequate

Table 4: Accepted extensions: preferred-model parameters. Radii in kpc; p, q window strengths ($p = 2 + M_N/M$). Parameter vector order: $S(M, R)$; $VW(M, R, r_v - R, M_N)$; $2L(M_1, R_1, M_2, R_2 - R_1)$; $VW2(M, R, r_{v1} - R, M_{N1}, r_{v2} - r_{v1}, M_{N2})$; $2LVW(M_1, R_1, M_2, R_2 - R_1, r_v - R_2, M_N)$; masses in $10^9 M_\odot$.

Galaxy	Model	k	Radii	Strengths	Parameters $\pm$ statistical error
NGC2841	VW2	6	R=1.20;r:v1=13.02;r:v2=44.35	p=14.3;q=-27.2	$10.5 \pm 2.2, 1.2 \pm 0.22, 11.8 \pm 0.85, 128 \pm 10, 31.3 \pm 2.1, -306 \pm 98$
NGC6015	2LVW	6	R1=0.72;R2=2.30;r:v=3.49	p=1.82	$0.896 \pm 0.077, 0.725 \pm 0.038, 7.21 \pm 0.63, 1.58 \pm 0.099, 1.19 \pm 0.19, -1.3 \pm 0.38$
NGC0300	VW	4	R=1.17;r:v=2.49	p=-6.52	$0.476 \pm 0.12, 1.17 \pm 0.16, 1.31 \pm 0.18, -4.05 \pm 0.3$
UGC00128	VW2	6	R=3.75;r:v1=13.53;r:v2=27.01	p=-3.02;q=3.6	$4.75 \pm 0.56, 3.75 \pm 0.32, 9.77 \pm 0.7, -23.8 \pm 1.4, 13.5 \pm 1.5, 7.62 \pm 5.1$
NGC1090	2LVW	6	R1=3.25;R2=3.26;r:v=10.93	p=3.86	$21.2 \pm 4.9e+07, 3.25 \pm 2.5e+06, 9.63 \pm 1, 0.0119 \pm 2.5e+06, 7.67 \pm 1.3, 17.9 \pm 4.2$
NGC7814	VW	4	R=0.38;r:v=0.93	p=4.61	$3.44 \pm 3.5e+02, 0.379 \pm 23, 0.553 \pm 43, 8.98 \pm 3.6$
NGC4138	VW	4	R=1.06;r:v=6.09	p=13.6	$3.62 \pm 3.6, 1.06 \pm 0.88, 5.03 \pm 1.6, 41.8 \pm 16$
UGC09133	VW2	6	R=0.06;r:v1=15.62;r:v2=70.42	p=307;q=2.21e+03	$0.374 \pm 0.02, 0.06 \pm 0.0031, 15.6 \pm 0.55, 114 \pm 5.7, 54.8 \pm 1.8, 825 \pm 2.1e+02$
UGC02487	VW2	6	R=0.68;r:v1=35.83;r:v2=53.12	p=105;q=13.3	$7.68 \pm 8.4, 0.685 \pm 0.74, 35.1 \pm 1.9, 794 \pm 1.6e+02, 17.3 \pm 3.7, 87.1 \pm 1.2e+02$
UGC03546	VW2	6	R=0.02;r:v1=0.61;r:v2=15.35	p=37.7;q=452	$0.104 \pm 3.9, 0.02 \pm 0.72, 0.594 \pm 0.97, 3.72 \pm 1.6, 14.7 \pm 1.5, 46.9 \pm 34$
NGC0289	VW2	6	R=0.02;r:v1=14.47;r:v2=24.28	p=1.38e+03;q=-2.91e+03	$0.0503 \pm 0.0044, 0.02 \pm 0.0019, 14.5 \pm 1.7, 69.4 \pm 28, 9.81 \pm 2.6, -146 \pm 32$
UGC06786	VW2	6	R=0.06;r:v1=1.88;r:v2=22.73	p=-47.8;q=517	$0.172 \pm 0.21, 0.0645 \pm 0.077, 1.81 \pm 0.18, -8.57 \pm 0.68, 20.9 \pm 0.88, 88.7 \pm 29$

Atlas: category-D fits with residuals

Category D: single-L (dotted), best model (solid; blue dashed = window radii, black dashed = Lagrangian reset), one-window fit (purple dashed). Residuals: single-L open, best filled.



Category D: single-L (dotted), best model (solid; blue dashed = window radii, black dashed = Lagrangian reset), one-window fit (purple dashed). Residuals: single-L open, best filled.

